

## BASIS FOR DECISION MEMO

Permit Processor: Bradley Popovich  
 Date: August 2, 2018  
 Permit No. MI0020257  
 Facility's Designated Name: Cadillac WWTP

Monitoring Point 001A: Authorization to discharge 3.2 MGD of treated municipal wastewater from Monitoring Point 001A through Outfall 001. Outfall 001 discharges to the Clam River.

|                                                             | <b>Maximum Limits for<br/>Quantity or Loading</b> |              |                 |                | <b>Maximum Limits for<br/>Quality or Concentration</b> |              |                 |              | <b>Monitoring<br/>Frequency</b> | <b>Sample<br/>Type</b>  | <b>Basis for<br/>Limits</b> |
|-------------------------------------------------------------|---------------------------------------------------|--------------|-----------------|----------------|--------------------------------------------------------|--------------|-----------------|--------------|---------------------------------|-------------------------|-----------------------------|
| <b>Parameter</b>                                            | <b>Monthly</b>                                    | <b>7-Day</b> | <b>Daily</b>    | <b>Units</b>   | <b>Monthly</b>                                         | <b>7-Day</b> | <b>Daily</b>    | <b>Units</b> |                                 |                         |                             |
| Flow                                                        | (report)                                          | ---          | (report)        | MGD            | ---                                                    | ---          | ---             | ---          | Daily                           | Report Total Daily Flow | PWJ                         |
| Carbonaceous Biochemical Oxygen Demand (CBOD <sub>5</sub> ) |                                                   |              |                 |                |                                                        |              |                 |              |                                 |                         |                             |
| <b>June – September</b><br><i>May – November</i>            | 190                                               | 270          | <b>(report)</b> | lbs/day        | 7                                                      | ---          | 10              | mg/l         | 5x Weekly                       | 24-Hr Composite         | WQBEL/AD                    |
| <b>October – November</b>                                   | <b>290</b>                                        | <b>480</b>   | <b>(report)</b> | <b>lbs/day</b> | <b>11</b>                                              | ---          | <b>17</b>       | <b>mg/l</b>  | <b>5x Weekly</b>                | <b>24-Hr Composite</b>  | PWJ                         |
| December – March                                            | 670                                               | 910          | <b>(report)</b> | lbs/day        | 25                                                     | ---          | 34              | mg/l         | 5x Weekly                       | 24-Hr Composite         | WQBEL/AD                    |
| <b>April – May / April</b>                                  | 450                                               | 670          | <b>(report)</b> | lbs/day        | 17                                                     | ---          | 25              | mg/l         | 5x Weekly                       | 24-Hr Composite         | WQBEL/AD                    |
| Total Suspended Solids (TSS)                                |                                                   |              |                 |                |                                                        |              |                 |              |                                 |                         |                             |
| <b>June – September</b><br><i>May – November</i>            | 530                                               | 800          | <b>(report)</b> | lbs/day        | 20                                                     | 30           | <b>(report)</b> | mg/l         | 5x Weekly                       | 24-Hr Composite         | PWJ                         |
| <b>October – May</b><br><i>December – April</i>             | 800                                               | 1200         | <b>(report)</b> | lbs/day        | 30                                                     | 45           | <b>(report)</b> | mg/l         | 5x Weekly                       | 24-Hr Composite         | PWJ                         |
| Ammonia Nitrogen (as N)                                     |                                                   |              |                 |                |                                                        |              |                 |              |                                 |                         |                             |
| <b>June – September</b><br><i>May – November</i>            | 24                                                | 53           | <b>(report)</b> | lbs/day        | 0.9                                                    | ---          | 2.0             | mg/l         | 5x Weekly                       | 24-Hr Composite         | WQBEL/AD                    |
| <b>October – November</b>                                   | <b>88</b>                                         | ---          | <b>(report)</b> | <b>lbs/day</b> | <b>1.6</b>                                             | ---          | <b>2.8</b>      | <b>mg/l</b>  | <b>5x Weekly</b>                | <b>24-Hr Composite</b>  | PWJ                         |
| December – March                                            | 99                                                | ---          | <b>(report)</b> | lbs/day        | 3.7                                                    | ---          | <b>(report)</b> | mg/l         | 5x Weekly                       | 24-Hr Composite         | WQBEL/AD                    |
| <b>April – May / April</b>                                  | 130                                               | ---          | <b>(report)</b> | lbs/day        | 4.8                                                    | ---          | <b>(report)</b> | mg/l         | 5x Weekly                       | 24-Hr Composite         | WQBEL/AD                    |
| Total Phosphorus (as P)                                     |                                                   |              |                 |                |                                                        |              |                 |              |                                 |                         |                             |
| May – March                                                 | 13                                                | ---          | <b>(report)</b> | lbs/day        | 0.5                                                    | ---          | <b>(report)</b> | mg/l         | 5x Weekly                       | 24-Hr Composite         | WQBEL                       |

| <u>Parameter</u>                  | <u>Monthly</u>              |     | <u>Daily</u> | <u>Units</u> | <u>Monthly</u>              |     | <u>Daily</u>           | <u>Units</u>    | <u>Monitoring Frequency</u> | <u>Sample Type</u> | <u>Basis for Limits</u> |
|-----------------------------------|-----------------------------|-----|--------------|--------------|-----------------------------|-----|------------------------|-----------------|-----------------------------|--------------------|-------------------------|
| April                             | 17                          | --- | (report)     | lbs/day      | 0.65                        | --- | (report)               | mg/l            | 5x Weekly                   | 24-Hr Composite    | WQBEL                   |
| Fecal Coliform Bacteria           | ---                         | --- | ---          | ---          | 200                         | 400 | (report)               | cts/100 ml      | 5x Weekly                   | Grab               | WQS                     |
| Perfluorooctane sulfonate (PFOS)  | (report)                    | --- | (report)     | lbs/day      | (report)                    | --- | (report)               | ng/l            | 2x Annually                 | Grab               | WQC                     |
| Perfluorooctanoic acid (PFOA)     | (report)                    | --- | (report)     | lbs/day      | (report)                    | --- | (report)               | ng/l            | 2x Annually                 | Grab               | WQC                     |
| Acute Toxicity                    | ---                         | --- | ---          | ---          | ---                         | --- | (report)               | TU <sub>A</sub> | Quarterly                   | 24-Hr Composite    |                         |
| Chronic Toxicity                  | ---                         | --- | ---          | ---          | 1.1                         | --- | ---                    | TU <sub>C</sub> | Quarterly                   | 24-Hr Composite    |                         |
| Available Cyanide                 | 0.16                        | --- | ---          | lbs/day      | 5.9                         | --- | (report)               | ug/l            | Monthly                     | Grab               |                         |
| Total Copper                      | 0.3                         | --- | 0.82         | lbs/day      | 11                          | --- | 31/(report)            | ug/l            | Weekly Quarterly            | 24-Hr Composite    | WQBEL                   |
| Total Nickel                      | (report)                    | --- | (report)     | lbs/day      | (report)                    | --- | (report)               | ug/l            | Quarterly                   | 24-Hr Composite    | WQBEL                   |
| Total Selenium                    | 0.15                        | --- | (report)     | lbs/day      | 5.7                         | --- | (report)               | ug/l            | Monthly                     | 24-Hr Composite    | WQBEL                   |
| Bis(2-ethylhexyl)phthalate        | 0.82                        | --- | (report)     | lbs/day      | 31                          | --- | (report)               | ug/l            | Monthly                     | 24-Hr Composite    | WQBEL                   |
| Total Mercury                     | (report)                    | --- | ---          | lbs/day      | (report)                    | --- | ---                    | ng/l            | Quarterly                   | Grab               |                         |
| Corrected                         | (report)                    | --- | (report)     | lbs/day      | (report)                    | --- | (report)               | ng/l            | Quarterly                   | Calculation        | WQC                     |
| Uncorrected                       | ---                         | --- | ---          | ---          | ---                         | --- | (report)               | ng/l            | Quarterly                   | Grab               | WQC                     |
| Field Duplicate                   | ---                         | --- | ---          | ---          | ---                         | --- | (report)               | ng/l            | Quarterly                   | Grab               | WQC                     |
| Field Blank                       | ---                         | --- | ---          | ---          | ---                         | --- | (report)               | ng/l            | Quarterly                   | Preparation        | WQC                     |
| Laboratory Method Blank           | ---                         | --- | ---          | ---          | ---                         | --- | (report)               | ng/l            | Quarterly                   | Preparation        | WQC                     |
|                                   | <u>12-Month Rolling Avg</u> |     |              |              | <u>12-Month Rolling Avg</u> |     |                        |                 |                             |                    |                         |
| Total Mercury                     | 0.000053                    | --- | ---          | lbs/day      | 2.0                         | --- | ---                    | ng/l            | Quarterly                   | Calculation        | WQV                     |
|                                   |                             |     |              |              | <u>Minimum % Monthly</u>    |     | <u>Minimum % Daily</u> |                 |                             |                    |                         |
| TSS Minimum % Removal             |                             |     |              |              |                             |     |                        |                 |                             |                    |                         |
| October – May<br>December – April | ---                         | --- | ---          | ---          | 85                          | --- | (report)               | %               | Monthly                     | Calculation        | STS                     |

| <u>Parameter</u> |     |     |     |     | <u>Minimum<br/>Daily</u> |     | <u>Maximum<br/>Daily</u> | <u>Units</u> | <u>Monitoring<br/>Frequency</u> | <u>Sample<br/>Type</u> | <u>Basis for<br/>Limits</u> |
|------------------|-----|-----|-----|-----|--------------------------|-----|--------------------------|--------------|---------------------------------|------------------------|-----------------------------|
| pH               | --- | --- | --- | --- | 6.5                      | --- | 9.0                      | S.U.         | <b>5x Weekly<br/>Daily</b>      | Grab                   | WQS                         |
| Dissolved Oxygen |     |     |     |     |                          |     |                          |              |                                 |                        |                             |
| May – November   | --- | --- | --- | --- | 6.0                      | --- | ---                      | mg/l         | 5x Weekly                       | Grab                   | WQBEL/AD                    |
| December – April | --- | --- | --- | --- | 5.0                      | --- | ---                      | mg/l         | 5x Weekly                       | Grab                   | WQBEL/AD                    |

**PERMIT CONDITIONS:**

Final Effluent Limitations, Monitoring Point 001A

**Quantification Levels and Analytical Methods for Selected Parameters**

Additional Monitoring Requirements

Pollutant Minimization Program for Total Mercury

**Pollutant Minimization and Source Evaluation Program for Perfluorooctane Sulfonate (PFOS) and/or Perfluorooctanoic Acid (PFOA)**

Untreated or Partially Treated Sewage Discharge Reporting and Testing Requirements

Facility Contact

Monthly Operating Reports

Asset Management

**Discharge Monitoring Report – Quality Assurance Study Program**

*Schedule of Compliance (not required)*

**Storm Water Pollution Prevention (not required)**

Michigan Industrial Pretreatment Program

Residuals Management Program for Land Application of Biosolids

**NOTES:**

The WQBEL Toxic memo recommended monitoring for the following parameters: 3,3-Dichlorobenzidine, Benzidine, Hexachlorobenzene, Hexachlorobutadiene, Hexachlorocyclopentadiene, Acrylonitrile, 4-Chloro-3-methylphenol, Fluoranthene, Pentachlorophenol, Phenanthrene, and 2,4,6-Trichlorophenol. These parameters have not been included in the draft permit because they are not expected in the discharge, however, the compounds and appropriate quantification levels (QL's) are now listed in Part I.A.2, and these parameters are included in the additional monitoring requirements under Part I.A.3.

Less stringent limits are recommended for Dissolved Oxygen for Monitoring Point 001A. Based on a review of the facility's existing effluent quality, Permits Section determined that it is appropriate to retain the current limits based on antidegradation.

Less stringent limits are recommended for CBOD<sub>5</sub>, and Ammonia Nitrogen (as N) for the months of December through September for Monitoring Point 001A. Based on a review of the facility's existing effluent quality, Permits Section determined that it is appropriate to retain the current limits based on antidegradation.

An analysis of the facility's existing data was conducted, and it was determined that 99% of the time the facility is meeting the daily PWJ limits for CBOD<sub>5</sub> and Ammonia Nitrogen (as N) for the months of October through November. The completed analysis also found that 95% of the time the facility is meeting the monthly PWJ limit for Ammonia Nitrogen (as N) for the months of October through November).

District Staff has approved a No Exposure Certification (NEC) on October 30, 2018, that will cover any storm water discharges during this permit reissuance cycle.

The requirement to report total mercury results and associated data for corrected, uncorrected, field duplicate, field blank, and laboratory method blank results is added in order to be consistent with the sampling and QA/QC requirements of EPA Method 1631, Revision E, *"Mercury in Water by Oxidation, Purge and Trap, and Cold Vapor Atomic Fluorescence Spectrometry"*.

Permits Section followed up with the facility and the facility confirmed that they are no longer using chlorine for disinfection.

The facility requested alternate seasons be evaluated within the WQBEL – conventional modeling due to weather factors within their northern geographical location. Alternate seasons were modeled and incorporated within the draft NPDES permit.

Limit Change Key

Normal Type = existing requirement - carried over from previous version of permit

**Bold Type** = new requirement - not in previous version of permit

*Italic* = deleted requirement - not carried over from previous version of permit

Basis for Limits Key

WQBEL - Water Quality-Based Effluent Limit

STS - Secondary Treatment Standard

WQC - Water Quality Concern

WQS - Water Quality Standard

WQV - Water Quality Variance

PWJ - Permit Writer's Judgment

AD – Antidegradation